

Robust Design of Perturb & Observe Maximum Power Point Tracking for Photovoltaic Systems

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Abstract—The perturb & observe (P&O) maximum power point tracking (MPPT) is widely used in photovoltaic (PV) systems due to its simplicity and reliability. To optimize the performance, a tradeoff between static and dynamic efficiencies should be made. Yet, present design methods are not efficient enough for practical use. As for the prior-art calculation-based methods, they only focus on the problem of direction confusion under dynamic conditions, while the static efficiency will be low. Although the experience-based methods consider both static and dynamic efficiencies, they cannot give a clear result, and the parameters still need to be set in a trial-and-error process. As such, in this paper, an analytical model is established based on the operation principles to provide clear characteristics of key indices (i.e., static and dynamic efficiencies, and the requirement of the DC voltage range), and the impact of parameters on the efficiency is revealed. Subsequently, a robust design is proposed for the optimum design, which can set the key parameters, achieving relatively high tracking accuracy. Finally, simulations and experimental tests are performed, verifying the theoretical analysis and the effectiveness of the proposed design.

Index Terms—Maximum power point tracking (MPPT), parameter design, perturb-and-observe (P&O), photovoltaic (PV) systems.

I. INTRODUCTION

THE photovoltaic (PV) system is highly based on power electronics, whose design process is always complex, and hence, it often needs to rely on the experience of engineers and trial-and-error methods. However, this approach is not scientifically rigorous and may not result in the most optimal design or provide theoretical guidance for system operation. To address these issues, an analytical model can be established to reveal the behaviors of the system. Consequently, the analytical model provides a scientific guideline for optimization and can be easily extended to other specifications.

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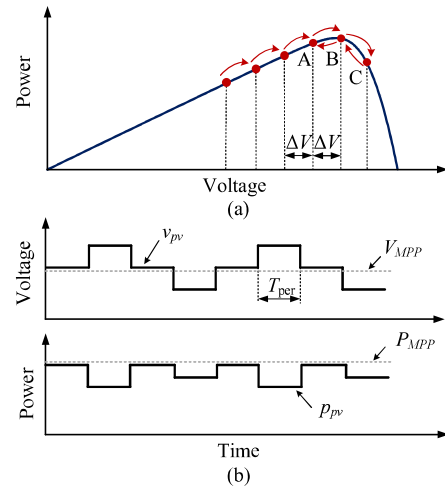


Fig. 1. Illustration of the P&O MPPT: (a) operation trajectories where the P&O MPPT periodically perturbs with a step of ΔV and finally oscillates between A, B and C and (b) steady-state waveform where the PV voltage v_{pv} and power p_{pv} oscillate around the corresponding voltage V_{MPP} and power P_{MPP} at the maximum power point (MPP). Here, T_{per} is the perturbation period ($T_{per} = 1/f_{per}$ with f_{per} being the perturbation frequency).

In PV systems, there is always a maximum power point tracking (MPPT) controller for energy harvesting [1]. As illustrated in Fig. 1(a), the power curve of PV panels always includes a maximum power point (MPP) that must be tracked accurately to achieve optimal efficiency [2], [3], [4], [5], [6]. To achieve high-performance MPP tracking, robustness is crucial to account for any potential challenges that may arise from a PV array, such as a rapid irradiance change. Additionally, high accuracy is necessary for efficient energy harvesting, since the efficiency of commercial products is at the level of 99.9% [7], [8], and simplicity is also important to release the stress of controllers and ensure high robustness. According to these, many algorithms for MPPT have been proposed [1], [2], [3], [4], [5], [6].

Basically, there are two types of MPPT algorithms: i) estimation type, which estimates the MPP based on its relationships with other points (e.g., open-circuit method [9], and artificial neural network (ANN) based method [10]), and it hardly achieves high efficiency due to nonideal characteristics of the PV panels. And ii) perturb-and-observe (P&O) type, which tracks the MPP by observing the variation by regular perturbation (such as dP -P&O [11], and adaptive P&O [12]), and it could reach an ultra-high efficiency. The P&O type has been widely used due to its high efficiency and robustness. Notably, the basic P&O

TABLE I
COMPARISON OF DESIGN METHODS FOR THE STEP SIZE

Type	Feature	Both the static and dynamic efficiency are considered	Conclusive result	Sufficient to determine the parameter
Experience-based	A range of ΔV [16]	YES	No	No
	A range of $\Delta V \times f_{\text{per}}$ [19]	YES	No	No
Calculation-based	Focus on the dynamic performance [18]	No	YES	No
	This work	YES	YES	YES

algorithm is still the most widely used MPPT algorithm due to the best simplicity [13], [14], [15], whose operating principle is shown in Fig. 1. There are two parameters of the P&O MPPT method that need to be designed, i.e., the perturbation frequency f_{per} and the step size ΔV . However, previous studies have not settled the design guideline well.

There are two key indices for benchmarking the P&O MPPT, i.e., the static and dynamic efficiencies, and both efficiencies should suit practical use. However, there is a tradeoff for the design of the step size ΔV between the indices. Although there are some design methods proposed [16], [17], [18], [19], all of which could not make the tradeoff well. As for the prior-art calculation-based design, [18] has pointed out that the step size should be large enough so that the problem of direction confusion under irradiance variations would not exist. However, it must consider the worst dynamic irradiance, leading to a very large step-size, ΔV . Hence, in [18], the static efficiency is not well considered, and it may be much lower than that of commercial products [7], [8]. In practice, the design of the step size is more experience-based [16], [19]. In [16], it is mentioned that modern MPPTs usually have a perturbation frequency higher than 10 Hz, whose step size should be smaller than 0.5% of its MPP voltage at standard test conditions (STC), $V_{\text{MPP_STC}}$. In [19], it is recommended that the perturbation speed ($\Delta V \times f_{\text{per}}$) should be in the range between 0.1% to 1% of $V_{\text{MPP_STC}}$ per second. In all, although the calculation-based method in [18] provides a specific parameter, the static efficiency would not be well considered. The experience-based methods in [16] and [19] could consider both static and dynamic efficiencies. However, these studies do not give a specific result, and engineers still need trial-and-error iterations to determine the parameters. Hence, all the existing design guidelines are not enough to determine the parameter of the step size suitable for practical use. To summarize, the analytical insight into all the key indices of the P&O MPPT has not been fully revealed, and naturally, it calls for a clear guideline to lead a performance-oriented design.

To fill this gap, firstly, this article reveals the characteristics of the key indices analytically. Then, a robust design for the P&O MPPT to balance its performances under static and dynamic irradiance conditions is proposed. The comparison of this work and other design methods for the step size of P&O MPPT is summarized in Table I. Compared to the conference paper [20], this article deepens the analysis with extensive experimental tests. The rest of this article is organized as follows. After describing the operation principles and corresponding characteristics of the MPPT in Section II, Section III presents the robust

design, which quantitatively explores the relationship between the performance and the parameters. And, the parameters are accurately selected with predefined performance indices. To verify the proposed analysis, Section IV shows the simulation results, and in addition, the comparison of the simulation and theoretical results is presented, verifying the effectiveness of the theoretical discussions. Finally, in Section V, experimental tests have been provided, which further validate the effectiveness of the proposed design. Concluding remarks are provided in Section VI.

II. PERFORMANCE OF THE P&O MPPT

For the P&O MPPT, it always operates under two conditions, i.e., static and dynamic irradiance. In the following, the corresponding characteristics under the two conditions are discussed, respectively.

A. Performance Under Static Irradiance

Fig. 1(a) presents the principle of the P&O MPPT. It periodically injects a fixed perturbation voltage of ΔV to the PV array, and then, observes the power variation of ΔP to determine the next perturbation direction. As such, it gradually moves to the MPP, as shown in Fig. 1(a). More specifically, the perturbation direction keeps when $\Delta P > 0$, and it is reversed if $\Delta P < 0$.

In the steady state, the PV system with the P&O MPPT will oscillate around three points, i.e., A, B, and C, as shown in Fig. 1(a). Accordingly, as observed in Fig. 1(b), the corresponding PV voltage v_{pv} has three levels in theory. Thus, the static efficiency can be obtained as

$$\eta = \frac{\eta_A + \eta_B + \eta_C}{3}, \quad \eta_i = \frac{p_{\text{pv}}(v_i)}{p_{\text{MPP}}}; \quad i = A, B, C \quad (1)$$

where v_i ($i = A, B, C$) indicates the voltage at point i , and $p_{\text{pv}}(v_i)$ is the power at point i , and p_{MPP} is the power at MPP. Then, the static efficiency η is the average of the efficiencies at points A, B, and C (i.e., η_A , η_B , and η_C).

In practice, the irradiance varies in a large range even in one day, and naturally, the static performance of MPPT algorithms should be evaluated considering different irradiance levels. According to EN50530 [21], two static efficiencies are usually adopted to evaluate the performance of MPPT methods, i.e., European Efficiency (η_{EUR}) and Californian Efficiency (η_{CEC}).

TABLE II
PARAMETERS OF THE PHOTOVOLTAIC ARRAY UNDER STC
(EN50530 STANDARD)

Parameter	Variable	Value
Maximum power	P_{MPP}	1000 W
Voltage at MPP	V_{MPP}	200 V
Open circuit voltage	V_{OC}	250 V
Short circuit current	I_{SC}	5.52 A

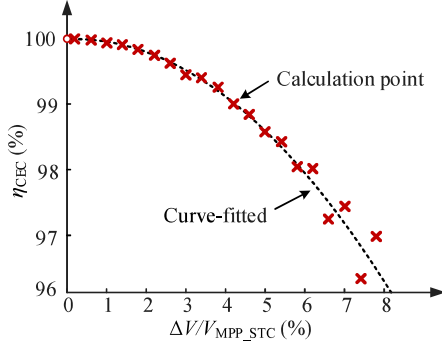


Fig. 2. Relationship between the Californian Efficiency η_{CEC} and the step size ΔV with the P&O MPPT algorithm. The voltage is normalized according to the voltage at the maximum power point under the standard test condition, V_{MPP_STC} . Notably, if ΔV is zero (being a constant voltage MPPT), its efficiency is not 100%.

In the following, η_{CEC} is taken as an example:

$$\eta_{CEC} = 0.04 \cdot \eta_{10\%} + 0.05 \cdot \eta_{20\%} + 0.12 \cdot \eta_{30\%} + 0.21 \cdot \eta_{50\%} + 0.53 \cdot \eta_{75\%} + 0.05 \cdot \eta_{100\%} \quad (2)$$

where $x\%$ ($x = 5, 10, 20, 30, 50, 100$) indicates the irradiance level is $x\%$ of the irradiance level at STC, that is 1000 W/m^2 at STC.

Notably, static efficiency is highly affected by ΔV . In order to investigate the impact of the perturbation step size, a PV array given in EN50530 with the P&O MPPT is exemplified. The PV array consists of 4×2 modules, and the parameters of the PV array are listed in Table II. Then, η_{CEC} is calculated, as shown in Fig. 2. According to Fig. 2, it is known that a larger step size will result in a lower static efficiency. It is worth mentioning that in this analysis, the PV voltage is normalized by V_{MPP_STC} .

B. Operation Under Dynamic Irradiance

In practice, dynamic irradiance profiles are common. Fig. 3 shows the condition where the irradiance G is increasing. If the MPPT takes a sample of the measured power at the instant k , i.e., P_k , then the next sample will be at the sampling instant $(k + 1)$, and $\Delta P = P_{k+1} - P_k$ is obtained to determine the next perturbation direction. Clearly, in addition to the power variation due to the perturbation, the irradiance change will also result in a power change. Hence, the total power difference is

$$\Delta P = \Delta P^{PO} + \Delta P^G \quad (3)$$

where ΔP^{PO} denotes the power variation caused by the perturbation, and ΔP^G indicates the power change due to the irradiance variation.

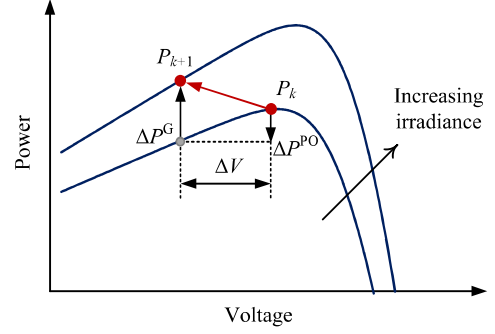


Fig. 3. After a perturbation period, PV power varies from P_k to P_{k+1} , and the variation is the sum of the power variations due to the perturbation ΔP^{PO} and the irradiance change ΔP^G . Here, ΔV indicates the step size.

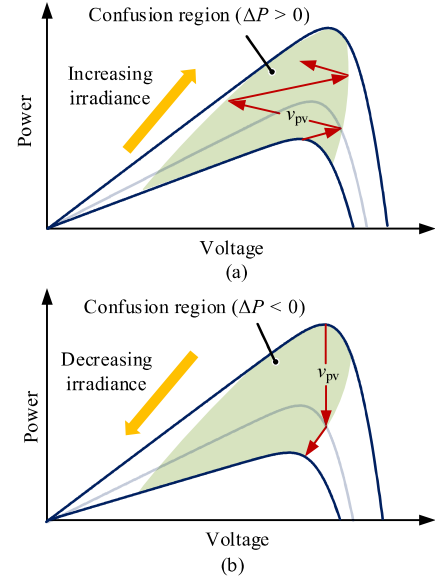


Fig. 4. Dynamic behaviors of the P&O MPPT algorithm under (a) an increasing irradiance condition, where the PV voltage v_{pv} remains going through the region of A_v (shaded region), and (b) a decreasing irradiance condition, where v_{pv} remains at the initial position until it reaches the boundary of A_v and moves along with it. The trajectories of the PV voltage are exemplified with red arrows.

Then, ΔP^G may make the P&O MPPT confused to determine the perturbation direction. Moreover, in the region of $|\Delta P^G| \geq |\Delta P^{PO}|$, the direction of the P&O MPPT may be confused, which is defined as A_v . In A_v , the sign of ΔP is identical to that of ΔP^G , but it is irrelevant to ΔP^{PO} . Specifically, when the irradiance increases, ΔP remains positive until it is out of the confusion region. Then, the direction reverses, and v_{pv} keeps going through the region, as shown in Fig. 4(a). Similarly, in the case of an irradiance decrease (see Fig. 4(b)), ΔP remains negative. Thus, the direction changes constantly until it reaches the boundary, and then, the operating point moves along with it, as illustrated in Fig. 4(b). Obviously, the above two dynamic incidents lead to a broader operating range and lower efficiency, which are not desired in PV systems.

C. Quantitative Analysis of Confusion Region A_v

As aforementioned, A_v is determined according to ΔP^G and ΔP^{PO} . To calculate A_v , it is necessary to derive the mathematical

expressions of the two power variations (i.e., ΔP^{PO} and ΔP^G). Firstly, it is known that irradiance directly affects the PV current. According to [1], there is

$$i_{pv} = I_n \frac{G}{G_n} - i_d \quad (4)$$

where I_n is the nominal photocurrent, G_n denotes the corresponding irradiance, and i_d is the diode current. Then, assuming that the irradiance has a change of ΔG after a perturbation, the PV current change is approximated as

$$\Delta i_{pv} = I_n \frac{\Delta G}{G_n} \quad (5)$$

with Δi_{pv} being the PV current variation after an irradiance change.

Notably, according to [2], the perturbation voltage is always less than 2% of V_{MPP_STC} , and then, the PV voltage is assumed unchanged during this process. Thus, ΔP^G can be simplified as

$$\Delta P^G = \frac{I_n}{G_n} v_{pv} \Delta G \quad (6)$$

Here, v_{pv} is the PV voltage. Additionally, because the perturbation is relatively fast (e.g., several tens of Hz), the irradiance change can be considered linear during a perturbation period, leading to

$$\Delta G = T_{per} \frac{dG}{dt} \quad (7)$$

Substituting (7) into (6), ΔP^G is derived as

$$\Delta P^G = \frac{I_n}{G_n} \frac{dG}{dt} T_{per} v_{pv} \quad (8)$$

Regarding ΔP^{PO} , as the perturbation is small, it can be approximated that the PV power, p_{pv} , varies linearly, thus,

$$\Delta P^{PO} = \Delta V \frac{dp_{pv}}{dv_{pv}} \quad (9)$$

Then, A_v can be quantitatively obtained:

$$\frac{I_n}{G_n} \frac{dG}{dt} T_{per} v_{pv} \geq \Delta V \left| \frac{dp_{pv}}{dv_{pv}} \right|, \quad v_{pv} \in A_v \quad (10)$$

From (10), it can be concluded that a larger step size ΔV leads to a narrower confusion region of A_v , which means higher efficiencies. A more detailed analysis is presented in Section III.

D. Discussion on Dynamic Performance

As shown in Section II, the dynamics may lead to a range of voltage variations due to direction confusion, and then, the dynamic efficiency is degraded. In practice, the voltage variation will widen the requirement for the operating voltage range of the PV inverter, meaning a higher stress for the controller. As a result, the dynamic efficiency is becoming an essential indicator. To enable a performance-oriented design, insights into the dynamics are to be revealed.

1) *Dynamic Evaluation in EN50530*: Firstly, before the performance-oriented design, the evaluation of the performance

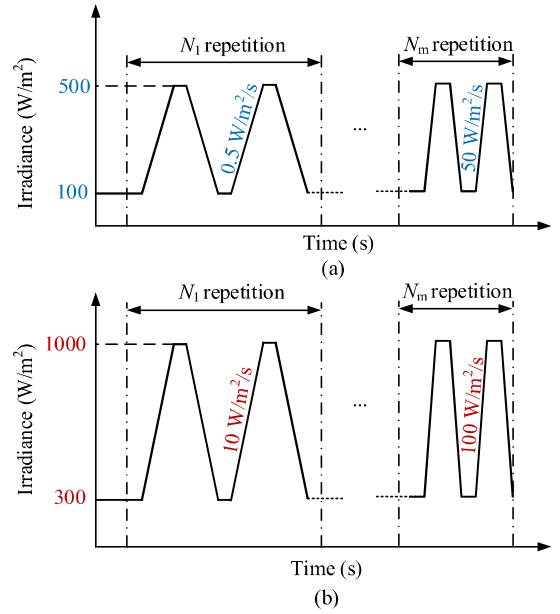


Fig. 5. Irradiance profiles in EN50530 to test the dynamic efficiency of the MPPT: (a) low irradiance profile and (b) high irradiance profile. Here, N_1 indicates the number of the first repetition, while N_m is the number of the m th repetition.

indices should be determined. In this paper, the EN50530 standard [21] is selected as an example, whose static efficiency is given as (2).

According to EN50530, two successive series of trapezoidal solar irradiance profiles should be used for the dynamic performance evaluation, as shown in Fig. 5, where each consists of several repetitive trapezoidal pulses with different ramp rates. Specifically, there is a waiting time zone before the repetition, and all of them have a dwelling time of 10 s after a rise to the top or down to the bottom [21], [22]. The first is for the case of low irradiance (100–500 W/m²), with ramp rates varying from 0.5 up to 50 W/m²/s, as denoted in Fig. 5(a). The other in Fig. 5(b) is for high irradiance (300–1000 W/m²), with ramp rates varying from 10 up to 100 W/m²/s. Then, the dynamic efficiency (defined as η_{dyn}) is calculated by the average of the efficiency along with the test profiles.

2) *Range of Voltage Variation*: The PV voltage varies when irradiance changes. The voltage variation range due to the irradiance dynamics is defined as $[V_{min}, V_{max}]$, being a challenge for both dynamic efficiency and system stability. To narrow the voltage range, the perturbation step size can be increased, while it, in turn, leads to a worse static efficiency, as discussed above. To properly design the step size, it is necessary to investigate the relation between $[V_{min}, V_{max}]$ and ΔV .

During irradiance increases, the PV voltage would regularly reach the boundary of A_v , as indicated in Fig. 4(a). To investigate the entire voltage variation, it is necessary to derive the corresponding irradiance values of V_{min} and V_{max} . For simplicity, the discrete perturbation is neglected, as it is always less than 2% of V_{MPP_STC} [2]. Hence, V_{min} and V_{max} are the lower and the upper boundary of A_v in the entire operation.

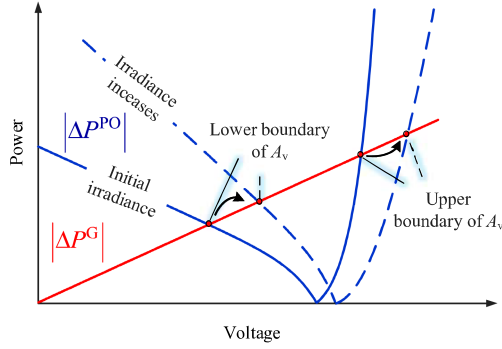


Fig. 6. Confusion region under an irradiance change, where the red represents the left part in (10) and the blue line is the right part, whose intersections form the boundary of the confusion region, A_v . The solid line is that under the initial irradiance while the dotted is that with stronger irradiance.

Firstly, it is obvious that the widest voltage range is under the steepest dynamic irradiance. Thus, only the irradiance profiles with the worst dynamics are considered, i.e., 100–500 W/m² with the rate of 50 W/m²/s and 300–1000 W/m² with the rate of 100 W/m²/s. Observing (10), only the right part relates to irradiance, where only the PV current i_{pv} is affected by the irradiance change. Hence, the power p_{pv} in (10) is substituted by the product of the current i_{pv} and voltage v_{pv} :

$$\left| \Delta V \frac{dp_{pv}}{dv_{pv}} \right| = \left| \Delta V \left(I_n \frac{G}{G_n} - i_d - \frac{di_d}{dv_{pv}} v_{pv} \right) \right| \quad (11)$$

from which, it can be concluded that, with a higher irradiance, $|\Delta P^{PO}|$ becomes larger when $v_{pv} = V_{min}$ (where dp_{pv}/dv_{pv} is a positive value) and it is to be smaller when $v_{pv} = V_{max}$ (where dp_{pv}/dv_{pv} is a negative value). Meanwhile, the MPP also moves right as the irradiance increases.

Based on (9) and (11), Fig. 6 presents how A_v (where $|\Delta P^{PO}| \geq |\Delta P^{PO}|$) changes under irradiance increasing, where $|\Delta P^{PO}|$ takes the MPP as the reference point and changes with the irradiance positively, and $|\Delta P^G|$ is independent of the irradiance and increases with the voltage proportionally. As shown in Fig. 6, when irradiance increases, the lower and higher boundaries of A_v move to the right.

To sum up, it is known that V_{min} is with the lowest irradiance under the dynamic irradiance of G_1 ($G_1 = 100 + 50t$ W/m²/s), that is, the first point reaching the lower boundary, where the irradiance is:

$$G = G_0 + \Delta G \frac{|V_{MPP} - v_{pv}|}{\Delta V} \quad (12)$$

Here, G_0 is the initial value of the dynamic irradiance, and V_{MPP} is the voltage at MPP under the irradiance of G_0 . Similarly, as shown in Fig. 6, V_{max} is with the highest irradiance under the dynamic irradiance of G_2 ($G_2 = 300 + 100t$ W/m²/s), that is, the last point reaching the upper boundary approximately, $G = 1000$ W/m².

Substituting the irradiance expressions into (10), the characteristic of the voltage variation range with perturbation frequency and step size can be obtained, as illustrated in Fig. 7. Furthermore, Fig. 8 presents the characteristic with step sizes under $f_{per} = 30$ Hz, which shows that V_{min} and V_{max} move

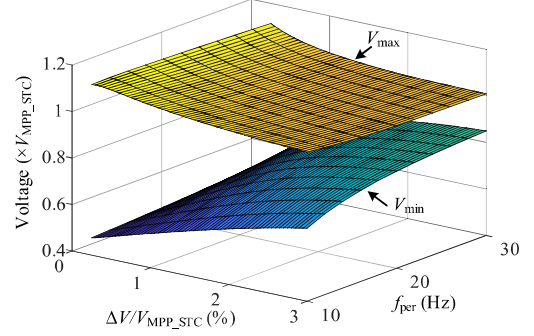


Fig. 7. Characteristic of the voltage variation range with the perturbation frequency f_{per} and step size ΔV . The yellow one represents the maximum voltage, while the blue is the minimum voltage.

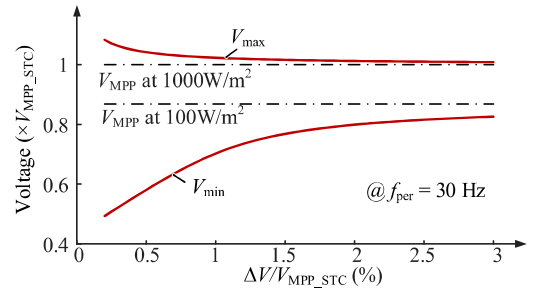


Fig. 8. Characteristic of the voltage variation range with the step size under the perturbation frequency f_{per} of 30 Hz. V_{max} denotes the upper boundary of the voltage range, while V_{min} being the lower boundary. With the increase of ΔV , V_{min} and V_{max} move closer to the MPP at 100 W/m² and 1000 W/m², respectively.

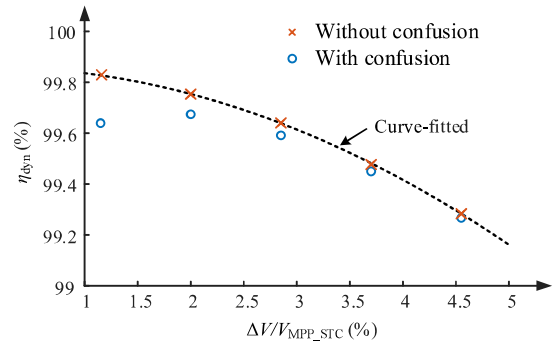


Fig. 9. Characteristics of the dynamic efficiency and its power loss due to direction confusion under different step sizes.

closer to the MPP at 100 W/m² and 1000 W/m² with the increase of ΔV , respectively.

3) *Dynamic Efficiency*: According to the previous discussion, there are power losses due to confusion under dynamics. To evaluate the impact on dynamic efficiency, an ideal P&O MPPT algorithm is considered without confusion, as the theoretical benchmark. Then, dynamic efficiencies under different step sizes were obtained through experimental tests, as shown in Fig. 9, where the ideal efficiency is only related to the step size (similar to the operating condition under static irradiance profiles). Hence, its characteristics can be obtained by curve fitting.

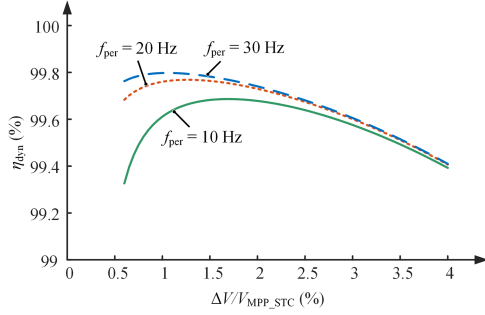


Fig. 10. Characteristics of the dynamic efficiency with the step size under different perturbation frequencies.

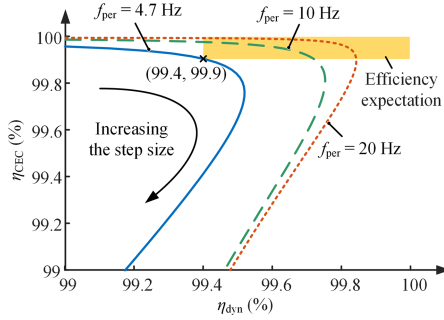


Fig. 11. Performance boundaries of the static and dynamic efficiency under different perturbation frequencies, from which, the perturbation frequency should be higher than 4.7 Hz to satisfy the efficiency expectations of more than 99.9% of static efficiency and more than 99.4% of dynamic efficiency.

It is obvious that the power loss due to confusion is directly related to the voltage variation, $[V_{\min}, V_{\max}]$. Based on the test points, its relationship can also be obtained, and as a result, the impact on the dynamic efficiency can be revealed analytically. Fig. 10 presents the characteristics of the dynamic efficiency with step sizes under different perturbation frequencies f_{per} .

Furthermore, since the characteristics of the static and dynamic efficiencies are revealed (cf., Figs. 2 and 10), the performance boundaries can be obtained. As shown in Fig. 11, the boundary is extended with the increase of the perturbation frequency. Notably, if the performance demand is out of the boundary, higher perturbation frequencies or other advanced MPPT techniques are required.

III. ROBUST DESIGN OF P&O MPPT

According to the theoretical analysis, a robust parameter design is presented in this section for the PV array referring to Table II, which is connected to a DC grid of 400 V.

A. Performance Expectations and Converter Design

Here, the performance expectations are determined first. In practice, the PV converter should operate in the MPPT voltage range (defined as $[V_{\text{MPPmin}}, V_{\text{MPPmax}}]$) to deliver the maximum power, and it should also ensure stability in the DC voltage range $[V_{\text{DCmin}}, V_{\text{DCmax}}]$ [28]. Accordingly, in this design, the range of the voltage variation is limited within the predefined voltage range, i.e., $[V_{\min},$

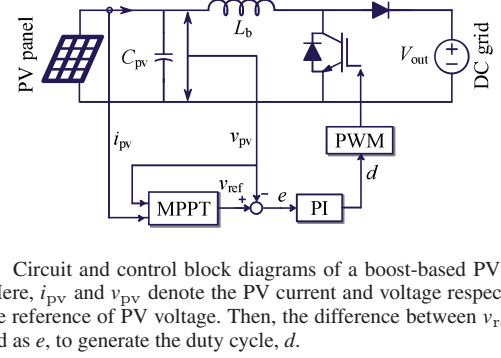


Fig. 12. Circuit and control block diagrams of a boost-based PV system to design. Here, i_{pv} and v_{pv} denote the PV current and voltage respectively, and v_{ref} is the reference of PV voltage. Then, the difference between v_{ref} and v_{pv} is denoted as e , to generate the duty cycle, d .

TABLE III
PARAMETERS OF THE BOOST CIRCUIT SHOWN IN FIG. 12

Parameter	Variable	Value
Input capacitor	C_{pv}	10 μF
Boost inductor	L_b	2.3 mH
Switching frequency	f_s	20 kHz
PI parameters	(k_p, k_i)	$(1 \times 10^{-6}, 4 \times 10^{-6})$
DC grid (load)	V_{out}	400 V

$V_{\max}] \in [V_{\text{DCmin}}/V_{\text{MPPmin}}, V_{\text{DCmax}}/V_{\text{MPPmax}}]$. Clearly, $[V_{\text{DCmin}}/V_{\text{MPPmin}}, V_{\text{DCmax}}/V_{\text{MPPmax}}]$ is different among PV converters. In this paper, without loss of generality, it is selected to be $[0.56, 1.1]$ as an example.

As for the dynamic efficiency, it should not be too small to limit the energy loss under dynamics, i.e., $\eta_{\text{dyn}} \geq \eta_{\text{dyn,min}}$. In this paper, the dynamic efficiency is expected to be more than 99.4%. Furthermore, considering the static efficiency of commercial MPPT systems is around 99.9% [7], [8], the designed parameters should also maintain a static efficiency of more than 99.9%. However, as shown in Fig. 11, the perturbation frequency challenges the tracking efficiency/performances of the P&O MPPT. Thus, the P&O MPPT should be designed to have a perturbation frequency over 4.7 Hz to satisfy the expectations of static and dynamic efficiencies.

Moreover, as shown in Fig. 11, a higher perturbation frequency means an improved performance. However, there are certain limitations, among which the most important one is that the settling time T_{Δ} of the rear-end converter should be smaller than the perturbation period T_{per} (i.e., $T_{\Delta} < T_{\text{per}}$) to ensure correct and stable operation. It means that the settling time of the power converter should be more than 220 ms, which is easy to implement even for the most conventional DC-DC converter. Based on the above discussions, a boost converter with a single-loop voltage control is selected for the boost converter (cf., Fig. 12), whose parameters are shown in Table III.

B. Perturbation Frequency

For the perturbation frequency, the main design principle is that the perturbation period T_{per} ($T_{\text{per}} = 1/f_{\text{per}}$) should be larger than the settling time T_{Δ} of the PV voltage after perturbation (i.e., $T_{\text{per}} > T_{\Delta}$), even in the worst case [23], [24], [25], [26].

With the design targets (performance expectations) presented in the last subsection, the maximum of T_{Δ} can be determined.

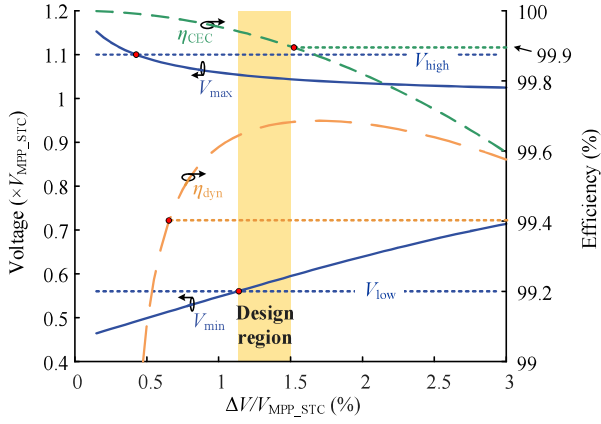


Fig. 13. Predicted performances with different step sizes under $f_{\text{per}} = 10$ Hz. Since the performance expectations are determined, the design region can then be obtained, satisfying the expectations.

In [26], it is pointed out that T_{Δ} reaches its maximum when the operating point is close to the MPP. Moreover, for PV panels connected to the boost converter, the worst case (seen from the perspective of the dynamic response) occurs when the panels operate at a low irradiance level [27]. Thus, the maximum of T_{Δ} is at the MPP under low irradiance levels. According to the above analysis, the maximum of T_{Δ} for the PV system shown in Fig. 12 is considered to be 70 ms according to extensive simulations. Then, $f_{\text{per}} = 10$ Hz (i.e., $T_{\text{per}} = 100$ ms $>$ 70 ms) is designed with a certain margin.

C. Perturbation Step Size

As shown in Figs. 2, 8, and 10, the MPPT performance highly depends on the perturbation step size. Since the quantitative relations are revealed, the performances can be well-designed. Subsequently, the design region of ΔV that satisfies the above constraints can be obtained, which is from 1.15% to 1.55% of $V_{\text{MPP_STC}}$, as summarized in Fig. 13. In this case, ΔV is selected to be 1.15% to ensure higher static efficiency, as shown in Fig. 13.

To summarize the design procedure, first, after the performance expectations are assigned, the task is shifted to designing the power electronics converter. Then, the perturbation frequency can be designed. After that, the performance characteristics of the step size are obtained. Finally, the design region can be obtained by combining the considered performance requirements. The selection of the step size is then easy to achieve, and its effectiveness can be verified by visually and analytically examining the corresponding resultant performances.

D. Design With Slower Perturbations

The analytical calculation not only provides a clear insight into the system design but also could be easily extended to other cases. As shown in Fig. 13, the design of the step size could fit the performance expectations well. Moreover, the proposed design for a different system is to be discussed. In this case, the power converter has a lower f_s of 10 kHz. Accordingly, considering the PV current ripple limits, L_b is changed to 4.6 mH. Then, f_{per} is modified to 5 Hz due to the slower dynamic response of the boost

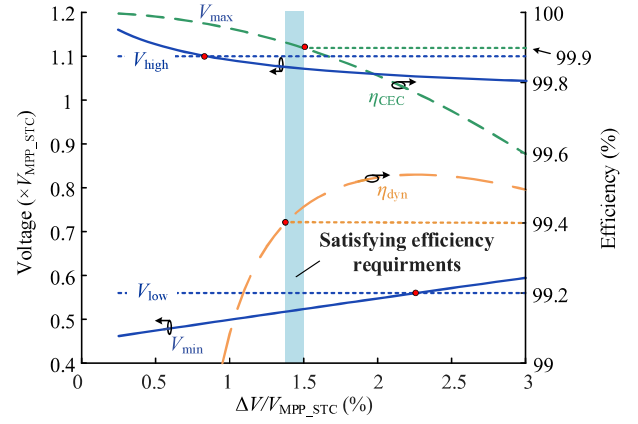


Fig. 14. Predicted performances with different step sizes under $f_{\text{per}} = 5$ Hz, where the switching frequency is 10 kHz.

converter. According to the proposed design, the performances are to be predicted, as shown in Fig. 14.

When comparing the design in Figs. 13 and 14, only the efficiency expectation is satisfied under the step sizes from 1.4% to 1.5% of $V_{\text{MPP_STC}}$ for the system mentioned in the last paragraph. The voltage range limitation could not be met at the same time. As such, in this case, the step size is selected to be 1.5% of $V_{\text{MPP_STC}}$, so that the voltage range can be as small as possible, according to Fig. 14.

Hence, the proposed design method could be easily extended to other cases. It provides a clear guideline for selecting parameters that balance well the key performances.

IV. SIMULATION RESULTS

In order to evaluate the proposed robust design and the above theoretical analysis, a MATLAB/Simulink model for the PV system in Fig. 12 was established. The PV panel and converter parameters are shown in Tables II and III.

A. Operation Demonstration

For the P&O MPPT parameters, initially, the designed results of $\Delta V = 1.15\% V_{\text{MPP_STC}}$ and $f_{\text{per}} = 10$ Hz are adopted. Then, comparisons with a large step size are performed, where the dynamic irradiance profiles are identical in both cases. Simulation results are shown in Figs. 15 and 16.

More specifically, Figs. 15(a) and 16(a), respectively, demonstrate the dynamic performance of the PV system under a low irradiance profile. By contrast, Figs. 15(b) and 16(b) show the performances under the high irradiance profile. As observed in Figs. 15 and 16, the theoretical predictions of the maximum and minimum voltages are in agreement with the simulation results. This proves that the theoretical analysis of the confusion region is accurate. Seen from Figs. 15(a) and 16(a), it is known that V_{min} is at the first point reaching the lower boundary under low dynamic irradiance, and the lower boundary becomes higher with the increase of the solar irradiance. Besides, V_{max} also appears in Figs. 15(b) and 16(b), where the irradiance dynamics are fast. The predictions are in consistency with the previous analysis. In addition, observations in Figs. 15 and 16 further confirm that

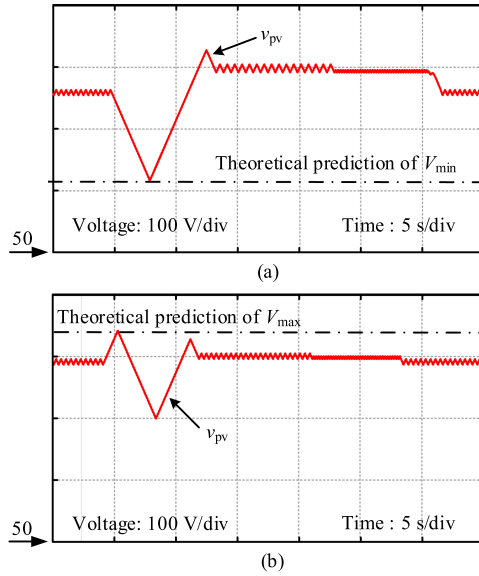


Fig. 15. Simulation results of the PV system with the designed parameters of $\Delta V = 1.15\%V_{MPP_STC}$ and $f_{per} = 10$ Hz under the dynamics of (a) 100–500 W/m² with the ramp rate of 50 W/m²/s, and (b) 300–1000 W/m² with the ramp rate of 100 W/m²/s. Here, v_{pv} is the PV voltage.

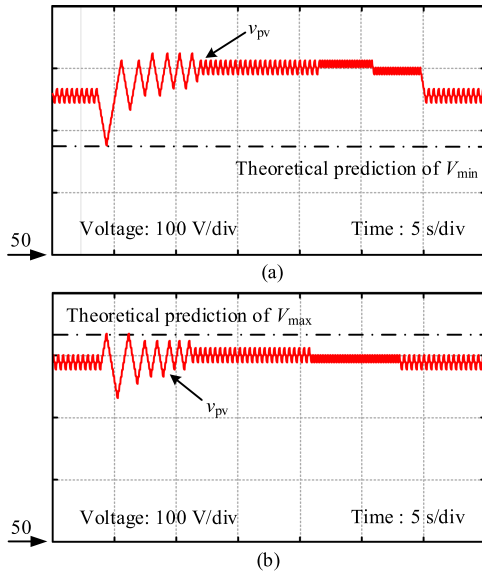


Fig. 16. Simulation results of the PV system with the designed parameters of $\Delta V = 2.85\%V_{MPP_STC}$ and $f_{per} = 10$ Hz under the dynamics of (a) 100–500 W/m² with the rate of 50 W/m²/s, and (b) 300–1000 W/m² with the rate of 100 W/m²/s.

the dynamics of the solar irradiance confuse the P&O MPPT. Eventually, the MPPT algorithm moves the operating point away from the real MPP, leading to increased power losses. This also agrees with the theoretical analysis and discussions in the previous sections.

B. Theoretical Verification

In order to confirm the theoretical model, a comparison was made between the theoretical prediction and the simulation results of the voltage variation range. The non-ideal factors of

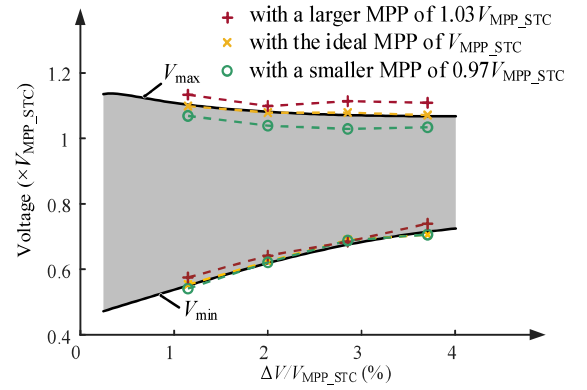


Fig. 17. Comparison of the theoretical prediction and the simulation results of the voltage variation range considering the discrete step. The simulation results of the ideal module with the MPP of V_{MPP_STC} , I_{MPP_STC} are noted as the cross. Considering uncertainties, the PV module with the MPP of $1.03V_{MPP_STC}$, $0.97I_{MPP_STC}$ is marked as the plus, and the other with the MPP of $0.97V_{MPP_STC}$, $1.03I_{MPP_STC}$ is marked as the circle.

the PV modules (e.g., manufacturing errors) were also taken into account. The simulation results of the ideal module with the MPP of V_{MPP_STC} , I_{MPP_STC} are presented in Fig. 17 and marked as crosses, agreeing with the theoretical prediction.

Besides, two cases of nonideal PV modules are also presented in Fig. 17. The first module has an MPP of $1.03V_{MPP_STC}$, $0.97I_{MPP_STC}$, marked with the plus sign. The second module has an MPP of $0.97V_{MPP_STC}$, $1.03I_{MPP_STC}$, marked with the circle. As shown in Fig. 17, the voltage variation range varies with MPP errors, which means that the non-ideal factors of the PV modules should also be taken into consideration for the DC voltage range.

V. EXPERIMENTAL VERIFICATION

To evaluate the proposed design and the analysis, experimental tests have been performed, referring to Fig. 12. Fig. 18(a) shows the experimental platform, while Fig. 18(b) illustrates the connections in the experimental setup used to establish the boost-based PV system. To assemble the power converter, Imperix silicon-carbide (SiC) power modules were utilized, while the control was implemented in an Imperix B-Box RCP. The boost converter is comprised of a standard PEB-SiC-8024 module and an inductor from the Imperix passives rack. To emulate the characteristics of PV panels, a PV simulator from ITECH was used, and a bidirectional DC power source was adopted as the DC grid (absorbing power), as seen in Fig. 18. Moreover, various variables and parameters can be monitored and changed in real-time with the help of the Imperix Cockpit.

A. Predication Performances

As mentioned in previous sections, the proposed design is based on the predicted performances by the theoretical model (e.g., dynamic efficiency, static efficiency, and the range of voltage variation). In this section, the accuracy of the theoretical predictions is to be verified by experimental tests. To verify the accuracy of predicting dynamic efficiency under any f_{per} , a perturbation frequency of 20 Hz (different from 10 Hz of the

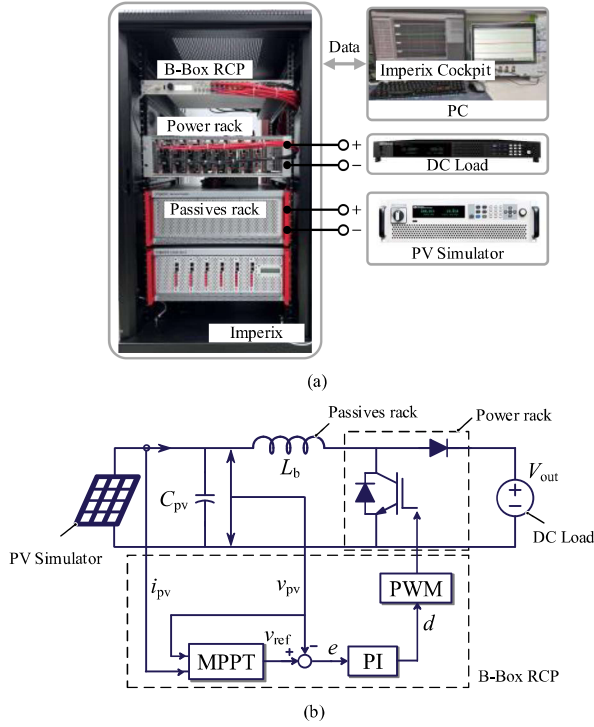


Fig. 18. Experimental setup of the PV system connected to a DC grid, where the DC load is a bidirectional DC source as the grid: (a) experimental platform, and (b) connections of the experimental setup to establish the boost-based PV system in Fig. 12.

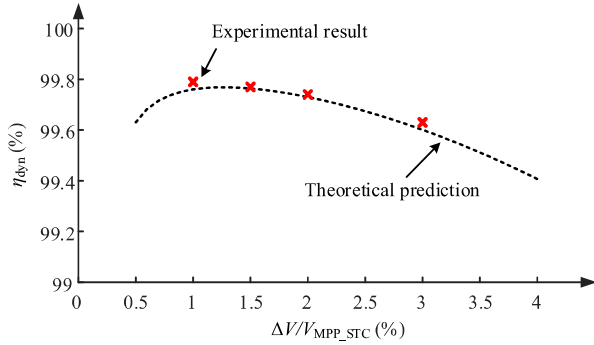


Fig. 19. Dynamic efficiency of the experimental tests and the theoretical prediction with different step sizes under $f_{\text{per}} = 20$ Hz.

analyzed in Fig. 9) is employed in the tests, where different step sizes are considered. The experimental results are shown in Fig. 19, which are also compared with the theoretical prediction results. It can be inferred from Fig. 19 that the prediction accuracy of the model is high.

With respect to the static efficiency, Fig. 20 presents the comparison between the experimental and theoretical results, where the difference is less than 0.3%, showing high agreement. Moreover, Fig. 21 illustrates the experimental results of the voltage range under different step sizes. As seen in Fig. 21, the difference between the experimental and theoretical results is less than one step size. To sum up, Figs. 19–21 have confirmed that the theoretical model is robust and accurate in predicting performances.

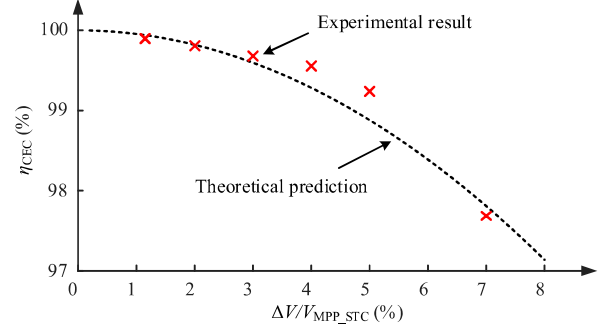


Fig. 20. Static efficiency of the experimental tests and the theoretical prediction with different step sizes under $f_{\text{per}} = 20$ Hz.

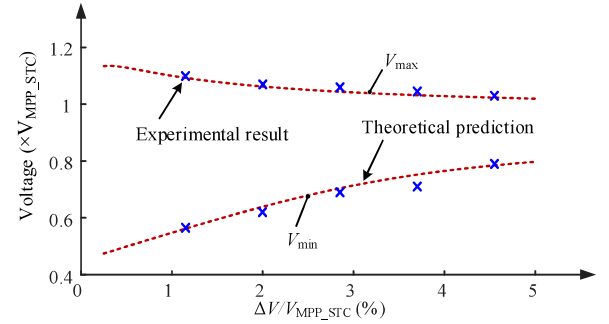


Fig. 21. Voltage ranges of the experimental tests and the theoretical prediction with different step sizes under $f_{\text{per}} = 10$ Hz.

B. Performance of the Proposed Design

The PV system with a perturbation frequency of 10 Hz and two design results of the step size are tested in order to verify the proposed design. According to the system parameters and the performance requirements, the step size of $\Delta V = 1.15\% V_{\text{MPP_STC}}$ was designed, as discussed in Section III. In addition, another set of the step size according to the design in [18], i.e., a large step size of $5\% V_{\text{MPP_STC}}$, was selected for comparison. The experimental test results under different irradiance dynamics are presented in Figs. 22–24.

In Fig. 22, the static performance of the two designed MPPT control algorithms is presented, which demonstrates the stable and accurate tracking of the active power under a constant irradiance level. Clearly, a larger step size will result in worse fluctuations, leading to a reduction in the overall static efficiency according to the analysis in Fig. 2. However, in the case of low dynamic irradiance (i.e., 100–500 W/m² with the ramp rate of 50 W/m²/s), as shown in Fig. 23, the P&O MPPT algorithm gets confused when the irradiance changes. Adoption of a large step size can improve the tracking performance to some extent, where the control rarely gets confused under irradiance dynamics, as discussed in [18]. The experimental tests shown in Fig. 23(b) confirm this. However, it should be pointed out that a large step size would incur larger power fluctuations when the system is going into a steady state. For instance, a close comparison of the PV voltage, v_{pv} , in Figs. 23 and 24, indicates the impact of the step size on voltage fluctuations, where the voltage variations become larger than those with a smaller step size.

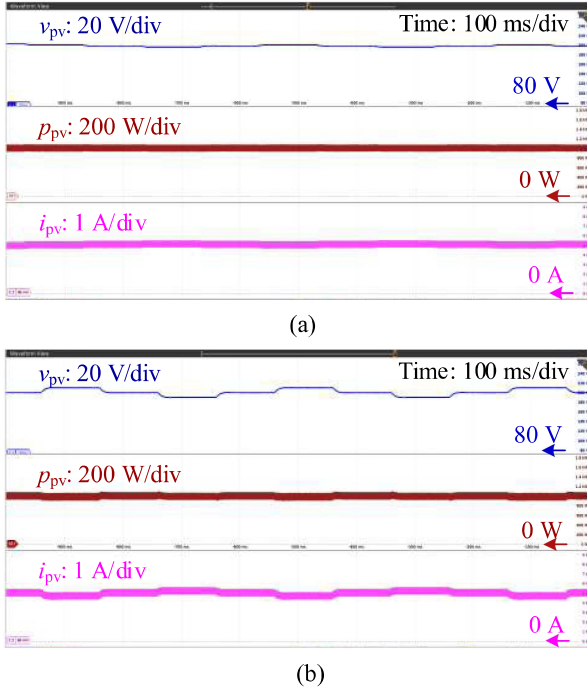


Fig. 22. Experimental results of the PV system with $f_{\text{per}} = 10$ Hz under the standard test condition: (a) $\Delta V = 1.15\%V_{\text{MPP_STC}}$ and (b) $\Delta V = 5\%V_{\text{MPP_STC}}$.

TABLE IV
PERFORMANCES COMPARISON OF THE DESIGN RESULTS UNDER THE
PERTURBATION FREQUENCY OF 10 HZ

Design Method	$\Delta V/V_{\text{MPP_STC}}$	η_{CEC}	η_{dyn}
Proposed	1.15%	99.90%	99.64%
Calculation-based in [18]	5%	99.26%	99.22%
Experience-based in [16]	0.2%	99.96%	98.39%

Furthermore, Fig. 24 shows the test results of the PV system under a fast-changing solar irradiance profile. When comparing the experimental results in Figs. 23 and 24, it can be inferred that a larger step size (shown in Figs. 23(b) and 24(b)), narrows the range of voltage variations during transients (i.e., irradiance is changing). And, with that, the maximum and the minimum voltages (i.e., V_{min} and V_{max}) are close to the voltages at the MPP under the solar irradiance levels being 100 W/m^2 and 1000 W/m^2 , respectively. This characteristic is also analyzed in Fig. 8, meaning that the experimental tests are in close agreement with the theoretical analysis.

In addition, Table IV demonstrates the performance comparison of three results by different design methods, which shows that the proposed design reaches a static efficiency of 99.90%, as expected. However, the efficiency of the PV system with the parameter set designed according to [18] is 99.26%, which results in about a 0.7% drop in efficiency. And, the dynamic efficiency of the proposed design is 99.64%, which is also superior to the prior-art design in [18] (being 99.22%). The discrepancy is because such a large step size leads to large power oscillations. According to the experience-based design in [16], i.e., a step size of 0.2%, a static efficiency 0.06% higher than the

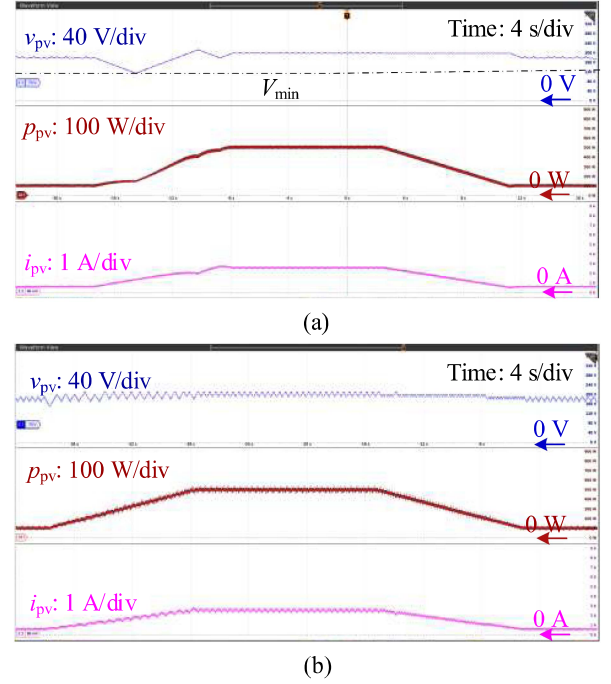


Fig. 23. Experimental test results of the PV system with $f_{\text{per}} = 10$ Hz under the dynamic irradiance changes of $100\text{--}500 \text{ W/m}^2$ with the rate of $50 \text{ W/m}^2/\text{s}$: (a) $\Delta V = 1.15\%V_{\text{MPP_STC}}$ and (b) $\Delta V = 5\%V_{\text{MPP_STC}}$.

TABLE V
PERFORMANCE COMPARISON OF THE DESIGN RESULTS UNDER THE
PERTURBATION FREQUENCY OF 5 HZ

Design Method	$\Delta V/V_{\text{MPP_STC}}$	η_{CEC}	η_{dyn}
Proposed	1.5%	99.85%	99.42%
Calculation-based in [18]	7%	97.70%	98.38%
Experience-based in [16]	0.2%	99.96%	96.43%

proposed design is achieved, however, the dynamic efficiency is degraded by about a 1.2% drop compared to the proposed design. Among these design methods, only the proposed design satisfies the performance expectations ($\eta_{\text{CEC}} \geq 99.9\%$, $\eta_{\text{dyn}} \geq 99.4\%$). In all, the predefined performance indices are properly considered in the proposed design.

As the efficiencies are excellent with the proposed design, furthermore, it could also provide specific requirements for optimizing the design of DC-DC converters, i.e., DC voltage range and dynamic response. With that, the proposed design could help to release the computational burden of the controller, making the design more cost-effective.

C. Performance With Slower Perturbations

For further verification, the boost-based PV system discussed in Fig. 12 is also tested experimentally. As discussed in Section III-B, in this case, f_{per} is modified to 5 Hz due to the slower dynamic response. With the test platform shown in Fig. 18, the design results of the step size by different methods are compared in Table V.

Based on the proposed design, a step size of 1.5% has been selected, resulting in a static efficiency of 99.85% and a dynamic

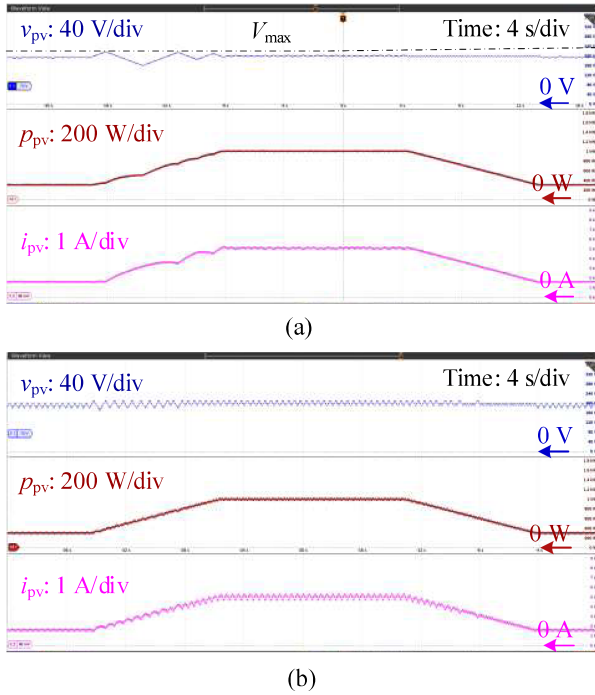


Fig. 24. Experimental test results of the PV system with $f_{per} = 10$ Hz under the dynamic irradiance changes of $300\text{--}1000\text{ W/m}^2$ with the rate of $100\text{ W/m}^2/\text{s}$: (a) $\Delta V = 1.15\%V_{MPP_STC}$ and (b) $\Delta V = 5\%V_{MPP_STC}$.

efficiency of 99.42%. Compared to the design in [18], both the static and dynamic efficiencies of the proposed design are higher by about 2% and 1%, respectively. Similar to that presented in Table IV, it necessitates designing the step size properly. Compared to the experience-based design in [16], although the proposed design has a static efficiency of about 0.1% lower, at the same time, it improves the dynamic efficiency by about 3%. All in all, the proposed design is much closer to the performance expectation ($\eta_{CEC} > 99.9\%$ and $\eta_{dyn} > 99.4\%$) than other methods. The experimental efficiency comparison in Table V further confirms that the proposed design method achieves better performances in terms of accurate predictions and comprehensive considerations, enabling the robust design of the PV system.

In conclusion, the proposed design outperforms the prior-art calculation-based design in [19] by effectively balancing static and dynamic efficiencies through accurate performance design. This is made possible by the high accuracy of the analytical model used. Compared to experience-based methods in [16] and [19], the proposed design provides clear guidelines for system design by directly selecting parameters in the visual performance plot. Additionally, the proposed design could be easily extended to other scenarios due to the high adaptability of the analytical model. Furthermore, more design parameters could be considered for system optimization. The proposed design optimizes both static and dynamic efficiencies while providing the requirements for DC-DC converter design, including dynamic response and DC voltage range.

In all, as presented in this paper, it is of importance to design the parameters of the MPPT algorithm considering the tradeoff

between static and dynamic performances under various irradiance profiles. Moreover, as discussed, an analytical model can be much beneficial to fully stimulate and evaluate the design in advance, avoiding costly repeating designs. The analytical models and the design method exhibit superior performances, as validated by simulations and experimental tests. The proposed robust design can be a promising guideline for practical PV systems with the P&O MPPT control.

VI. CONCLUSION

In order to balance the static and dynamic efficiencies as well as the requirement of the DC voltage range of the P&O MPPT system, in this paper, an analytical model was established to reveal the behaviors of the P&O MPPT operation. Firstly, the characteristic of static efficiency was derived according to the three-level operation. As for the dynamic efficiency and range of voltage variations, those were determined by direction confusion. Hence, the confusion region was analyzed first, and then the performance characteristics could be obtained. Additionally, curve fitting was utilized to simplify mathematical relationships. After the analytical model was established, a robust design guideline was presented to achieve the optimal design. From the DC-DC converter to P&O MPPT parameters, all the design considerations are optimized and satisfy the performance expectations well. Simulations and experimental tests have validated the accuracy of the model and the effectiveness of the proposed robust design.

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